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Analysis with sodium carbonate extract

Preparation of sodium carbonate extract:

Take 1g of the given salt and 3g of solid sodium carbonate in a100mL beaker. Add 20g of distilled water to it. Heat the beaker with its contents on a hot plate or Bunsen burner. After boiling the solution for few mins, filter it through a filter paper in a funnel and collect the filtrate in an another beaker. The filtrate is called sodium carbonate extract.

10.	Test for halides: To about one mL of the sodium carbonate extract add dil. HNO_3 in drops with shaking until the effervescence ceases, and then add about 1mL of $AgNO_3$, and shake it well.	A curdy white precipitate(ppt) insoluble in about 1mL of dil. ammonia is formed	Presence of chloride
		A pale yellow ppt sparingly soluble in ammonia is formed	Presence oh bromide
		A black ppt is formed	Presence of sulphide
11	Test with barium chloride: To about one mL of the sodium carbonate extract, add dil. acetic acid in drops with shaking until the effervescence ceases, then add 1mL of barium chloride solution and shake it.	A white ppt is formed insoluble in dil H_2SO_4	Presence of sulphate
12	Test with lead acetate: To about 1mL of the sodium carbonate extract, add 1mL of dil acetic acid and heat it , until the effervescence ceases, and then add 1mL of lead acetate	A whit ppt soluble in excess of ammonium acetate is formed	Presence of sulphate
13	Brown ring test: To about 1mL o f the sodium carbonate extract add dil. H_2SO_4 in drops with shaking until the effervescence ceases and about 0.5mL of freshly prepared ferrous sulphate solution. Then keeping the test tube in a slanting position add Conc. H_2SO_4 along the sides of the test tube.	A brown ring is formed	Presence of nitrate